Git behind the scenes

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Git behind the scenes

Git is a version control system that allows you to track changes to your files and folders. It is a powerful tool that can help you manage your code more effectively. In this section, we will explore the basics of how git works internally.

Git Snapshots

A git snapshot is a point in time in the history of your code. It represents a specific version of your code, including all the files and folders that were

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A git snapshot is a point in time in the history of your code. It represents a specific version of your code, including all the files and folders that were present at that time. Each snapshot is identified by a unique hash code, which is a string of characters that represents the contents of the snapshot.

A snapshot is not an image, it's just a representation of the code at a specific point in time. Snapshot is a loose term that is used when git stores information about the code in a locally stored key-value based database. Everything is stored as an object and each object is identified by a unique hash code.

3 Musketeers of git

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- Commit Object
- Tree Object
- Blob Object

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Commit Object

Each commit in the project is stored in `.git` folder in the form of a commit object. A commit object contains the following information:

- Tree Object
- Parent Commit Object
- Author
- Committer
- Commit Message



Tree Object

Tree Object is a container for all the files and

Tree Object

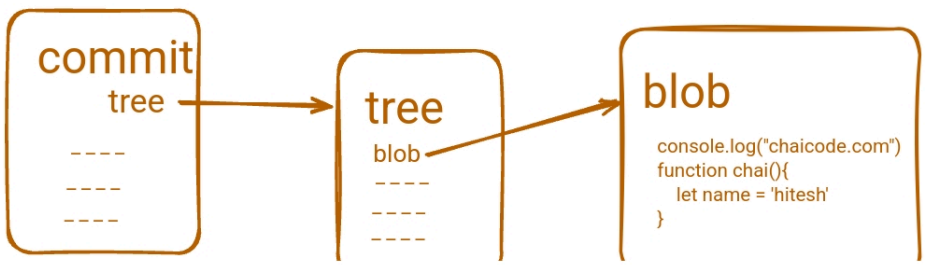
Tree Object is a container for all the files and folders in the project. It contains the following information:

- File Mode
- File Name
- File Hash
- Parent Tree Object

Everything is stored as key-value pairs in the tree object. The key is the file name and the value is the file hash.

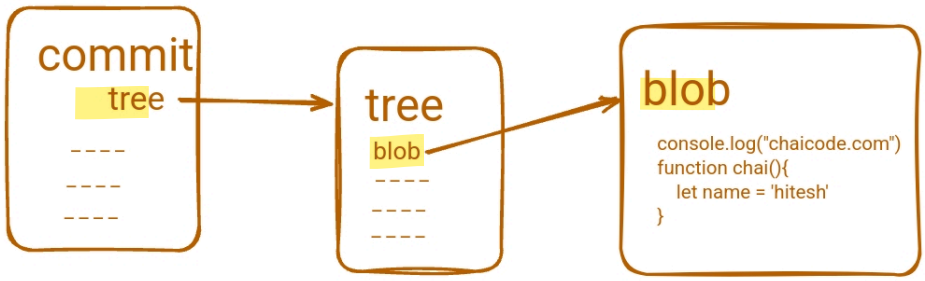
Blob Object

Blob Object is present in the tree object and contains the actual file content. This is the place where the file content is stored.



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Helpfil commands

Here are some helpful commands that you can use to explore the git internals:

```
git show -s --pretty=raw <commit-hash>
```

Grab tree id from the above command and use it in the following command to get the tree object:

```
git ls-tree <tree-id>
```

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```
git ls-tree <tree-id>
```

Grab tree id from the above command and use it in the following command to get the blob object:

```
git show <blob-id>
```

Grab tree id from the above command and use it in the following command to get the commit object:

```
git cat-file -p <commit-id>
```

Previous
Terminology